
Engineering PORTFOLIO

CAD Design | Prototyping | Manufacturing | Problem Solving



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PROJECTS

Selected Work

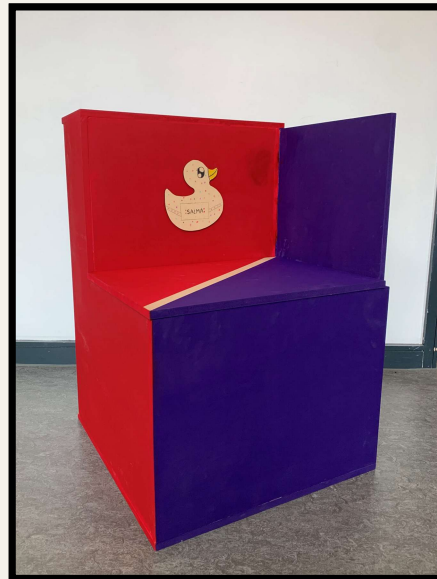
Four projects presented with image-led case study pages.



PROJECT 1

PS4 Gaming Briefcase (Individual)

CAD / Prototype / Manufacture / Laser cutting / Electronics



PROJECT 2

Multifunctional Chair (Individual)

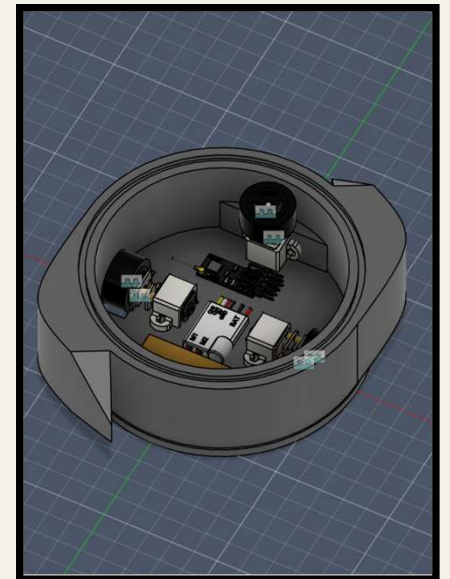
Scaled Modelling / Manufacture / Laser Cutting / Joint Techniques



PROJECT 3

Race Car Scrap Competition (Group)

Problem Solving / Soldering / Electronics / Team-work



PROJECT 4

Battle Bot (Current Project)

CAD / Prototype / Manufacture / Electronics / Soldering

PS4 Gaming Briefcase



Portable console setup | Built-in screen | Built in fan cooler | Plug & Play | Controller holder

GOAL

To design and manufacture a portable gaming briefcase that allows a PlayStation 4 to be transported safely and used without needing a separate TV or monitor. The product was aimed at console gamers who want a quicker, more convenient setup when gaming away from home. The briefcase needed to combine protection, storage, ventilation, and a dark technology-inspired aesthetic.

CHALLENGES

The main challenge was creating a compact internal layout that could securely fit the PS4, screen, controller, cables, and ventilation features while remaining practical to carry and safe to use. Key issues included heat management, cable organisation, material strength, electronic protection, and keeping the design compact without making the product feel overcrowded. Another challenge was figuring out how to disassemble an old laptop and salvage the screen.

SOLUTIONS / RESULTS

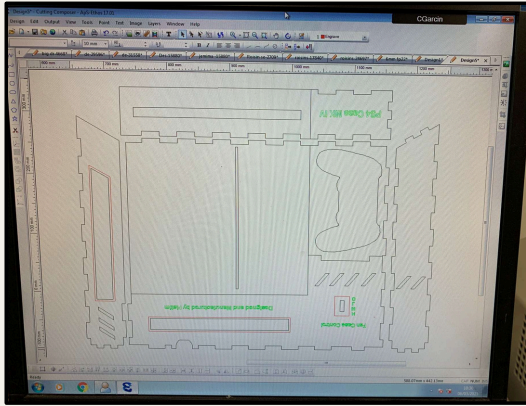
The final prototype used a plywood briefcase structure with a built-in salvaged laptop screen, internal accessory storage, ventilation openings, and secure mounting points for the console and components. CAD modelling was used to plan the layout before manufacture, helping refine the proportions, screen position, and internal spacing. The finished product reduced setup time, improved portability, protected the console during transport, and achieved the client's preferred dark, modern gaming style.

Skills: CAD Design | Electronics | Prototyping | Manufacturing | Material Testing | Client Research | Problem Solving | Joining Techniques | Sanding and Finishing | Painting | Laser Cutting

PROJECT 1 – INDIVIDUAL PROJECT

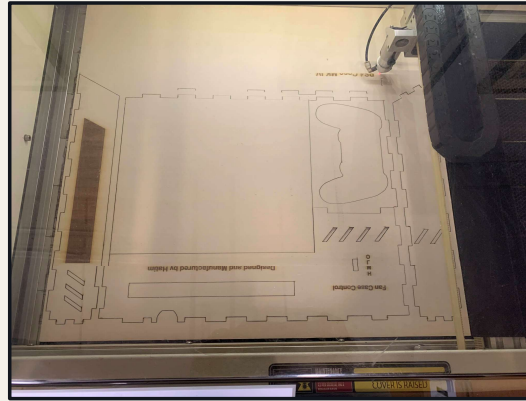
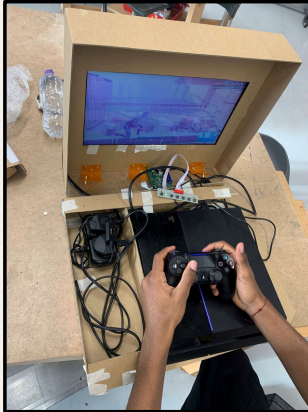
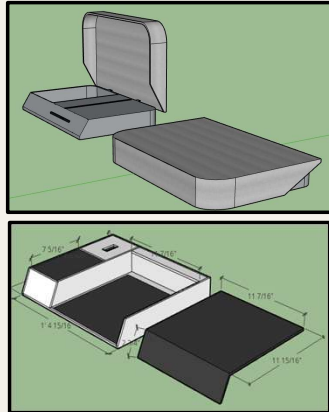
Development Evidence

CAD, making process and final details.



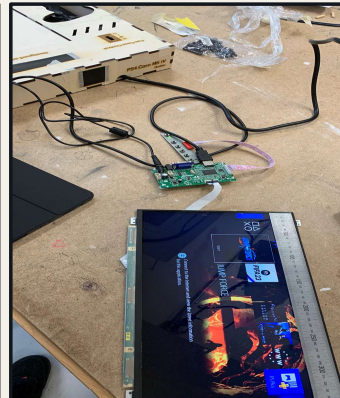
CAD LAYOUT

Used to plan the internal layout, screen position, console location and storage areas.



MANUFACTURING

Cutting, sanding, drilling, assembly and finishing were used to produce the plywood prototype. Screen from a laptop was also calibrated and tested during this process.



FINAL DETAILS

Ventilation, controller storage and cable routing helped improve usability and safety.



Multifunctional Chair



Built-in storage space | Removable seat | Hinged access door | Strong plywood structure

GOAL

The goal was to design and manufacture a multifunctional chair for my client, my sister. The chair needed to work as both a seat and a convenient storage unit, helping save space while keeping items organised and easy to access. Instead of making a chair with only one purpose, I wanted the final design to be practical, compact, and useful for everyday storage.

CHALLENGES

The main challenge was making sure the chair had enough storage space while still being strong and comfortable to use. A difficult part of the making process was fitting the hinged door correctly, as it needed to open smoothly without being too tight or swinging open too loosely. I also had to adjust the design after client feedback, removing some access below the seat so items could be stacked vertically without falling to the side.

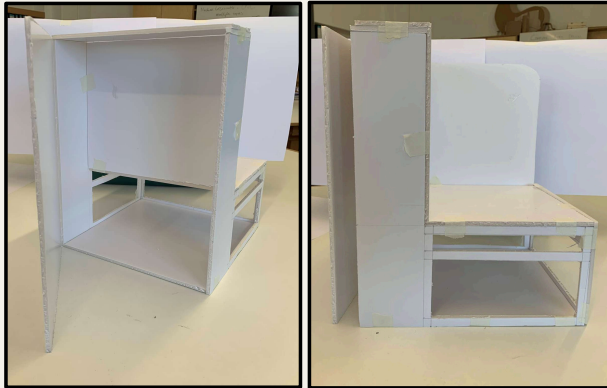
SOLUTIONS / RESULTS

The final product was made mainly from plywood, using dowel joints, wood glue, screws, hinges, wood filler, sanding, filing, painting, and a laser-cut logo to complete the design. The jointing method helped make the structure stronger, while the hinged door allowed the storage area to be accessed more easily. My client was happy with the result because her feedback was included throughout the process, and the chair successfully combined seating with practical storage.

Skills: Client Research | Product Design | Prototyping | Woodworking | Jointing Techniques | Sanding and Finishing | Painting | Laser Cutting | Problem Solving | Manufacturing

Development Evidence

CAD, making process and final details.



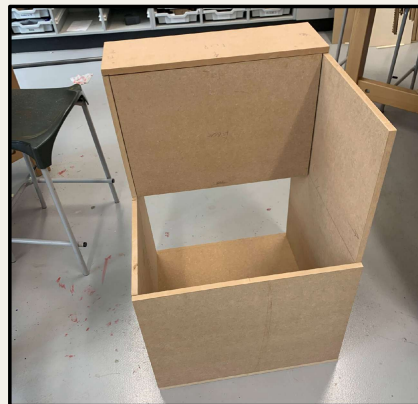
MODEL LAYOUT

The foam paper model was used to test the shape, size, and opening design before building the full chair.



MANUFACTURING

The plywood pieces were measured, cut, sanded, and assembled to create the main cabinet structure.



FINAL DETAILS

The final design was painted red and purple, with a hinged door, storage space, ventilation, and cable access added for function and appearance.



Race Car Scrap Competition



Custom-built chassis | Front rubber drive wheels | Electronics | Compact low-profile body | Easy to drive

GOAL

The goal was to design and build a race bot for a team competition (Electronics and Amateur Radio Society) using scrap materials and limited resources. The bot needed to work on the circuit and perform as well as possible, even though we had limited tools, space, and time. We focused on creating a lightweight and functional design that could move reliably and compete against the other teams.

CHALLENGES

The main challenge was making the bot work with the materials we had available, especially because most of the body was made from scrap plastic containers. The plastic was difficult to cut accurately, and the small adjustments took time because the workspace and tools were limited. We also had to change the original design from a fully enclosed box shape to a more Cybertruck-style body with gaps, which saved time and made the bot easier to assemble.

SOLUTIONS / RESULTS

The final race bot included basic RC electronics, soldered connections, and a lightweight plastic body made from reused materials. As a team, we solved problems during the build by adjusting the shape, making small cuts, and changing the design so it could be completed in time. The bot performed well in the competition and our team came second, although we found that the wheels needed more grip to get over the ramp more easily.

Skills: Teamwork | Soldering | Electronics | Problem Solving | Prototyping | Material Testing | Cutting and Adjusting | Time Management | Design Development | Competition Engineering

PROJECT 3 – GROUP PROJECT

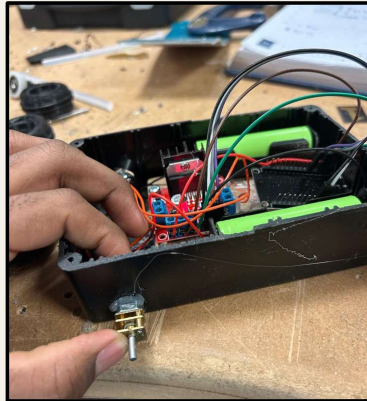
Development Evidence

CAD, making process and final details.



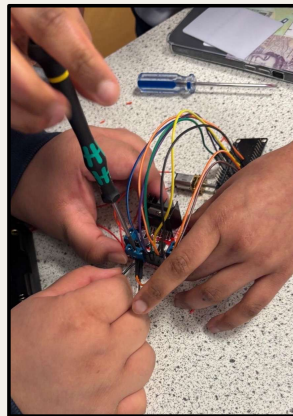
GENERAL PICTURES

Scouting the competition track early helped us plan for the tight curves, sharp bumps, and long straights that would define our car's design and features.



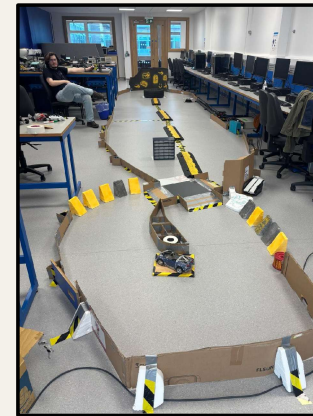
MANUFACTURING

Using soldering irons, sanders, and cutting tools, the team handcrafted every component from raw materials to a fully working vehicle.



FINAL RESULTS

Finishing in second place, our car completed two full laps, handled most collisions without damage, and remained smooth and responsive throughout the entire run.



CONTACT

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Final Skills Summary

Design: Sketching | Design Development | User Research | Evaluation
CAD / Digital: 3D Modelling | Rendering | Technical Layouts | Electronics
Manufacturing: Cutting | Sanding | Drilling | Assembly | Finishing
Professional: Problem Solving | Time Management | Attention to Detail

Thank you



"Work experience at HS2, 2022"



"Presenting PS4 briefcase at open day."